



ANNA M. ROMANOVSKAYA, PhD

Research Scientist | Stellar Spectroscopy | Binary Stars | Machine Learning

CONTACT

annarom@inasan.ru
Institute of Astronomy,
Russian Academy of Sciences
Moscow, Russia

LINKS

[linkedin.com/in/astroram](https://www.linkedin.com/in/astroram)
[Google Scholar](#)

RESEARCH IDS

ORCID

0000-0002-9879-6822

Web of Science ResearcherID

D-3111-2019

Scopus Author ID

57194060399

CORE SKILLS

Spectroscopic analysis
Python
Data analysis
Numerical optimization
Machine learning methods

LANGUAGES

English

Professional working

Spanish

Elementary

RESEARCH PROFILE

Research scientist specializing in high-resolution stellar spectroscopy and stellar atmosphere modelling. Experienced in determining atmospheric parameters and chemical abundances in magnetic chemically peculiar stars, metallic-lined stars, normal A-B stars, and spectroscopic binary systems. Current research focuses on hybrid approaches combining classical spectral synthesis, chi-square optimization, and machine learning for the analysis of composite spectra of binary stars.

ACADEMIC APPOINTMENTS

2025-Present Scientific Researcher

Institute of Astronomy, Russian Academy of Sciences

2017-2025 Junior Researcher

Institute of Astronomy, Russian Academy of Sciences

2015-2017 Research Assistant

Institute of Astronomy, Russian Academy of Sciences

EDUCATION

2025-2027 Yandex School of Data Analysis

Artificial Intelligence in Natural Sciences (professional program)

2017-2021 PhD in Astronomy and Astrophysics

Institute of Astronomy, Russian Academy of Sciences

Degree awarded: 2022

2011-2016 Specialist's degree in Astronomy and Astrophysics

Lomonosov Moscow State University

RESEARCH INTERESTS

- High-resolution spectroscopy of chemically peculiar Ap/Bp stars, Am stars, normal A-B stars, and spectroscopic binary systems.
- Stellar atmosphere modelling, spectral synthesis, LTE/non-LTE abundance analysis, vertical stratification and SED fitting.
- Machine-learning and numerical-optimization methods for parameter determination in SB2 composite spectra.

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METHODS

Research areas

Stellar spectroscopy; Ap/Bp, Am and normal A-B stars; spectroscopic binaries

Methods

Spectral synthesis; LTE/non-LTE abundance analysis; SED fitting; chi-square optimization; PCA; ML regression

Software

Python, NumPy, SciPy, pandas, scikit-learn, LightGBM, XGBoost, PyTorch, Optuna

PUBLICATION METRICS

15 peer-reviewed publications since 2021

15 indexed in RSCI / WoS / Scopus

8 publications in Q1 journals

GRANTS, FUNDING AND PUBLICATIONS

Academic funding and selected peer-reviewed publications

GRANTS AND FUNDING

- 2026-2028 - Russian Science Foundation, Project No. 26-72-00007, Hybrid method for parameter determination of spectroscopic binary stars based on spectral synthesis and machine learning. Role: Principal Investigator. Status: submitted / under review.
- 2024-2025 - Russian Science Foundation, Project No. 24-22-00237. Role: project participant.
- 2022-2023 - Grant of the President of the Russian Federation for young scientists, Project No. MK-5127.2022.1.2. Role: project participant.
- 2019-2022 - Russian Foundation for Basic Research, Project No. 19-31-27001, Atmospheric Parameters of Magnetic Chemically Peculiar Stars. Role: project participant.

SELECTED PUBLICATIONS

15 peer-reviewed publications since 2021; 15 indexed in RSCI / WoS / Scopus; 8 publications in Q1 journals.

1. **Romanovskaya, A., & Zvyagintsev, S.** Abundance Analysis of the Spectroscopic Binary α Equulei. *Galaxies*, 13(4), 88, 2025.
2. **Romanovskaya, A. M., Ryabchikova, T. A., Mashonkina, L. I., Sitnova, T. M., & Serebryanova, N.** Non-LTE abundance analysis of A-B main-sequence stars. III. Criteria for normal stars. *Astronomy Letters*, 51(1), 13-27, 2025. DOI: 10.31857/S0320010825010022.
3. **Romanovskaya, A., Ryabchikova, T., Pakhomov, Yu., Potravnov, I., & Sitnova, T.** Slowly Rotating Peculiar Star BD+00°1659 as a Benchmark for Stratification Studies in Ap/Bp Stars. *Galaxies*, 12(5), 55, 2024.
4. **Romanovskaya, A. M., Ryabchikova, T. A., Pakhomov, Yu. V., Korotin, S. A., & Sitnova, T. M.** Non-LTE abundance analysis of A-B stars with low rotational velocities - II. Do A-B stars with normal abundances exist? *Monthly Notices of the Royal Astronomical Society*, 526, 3386, 2023. DOI: 10.1093/mnras/stad2862.
5. **Romanovskaya, A. M., Shulyak, D. V., Ryabchikova, T. A., & Sitnova, T. M.** Fundamental parameters of Ap-stars GO And, 84 UMa and κ Psc. *Astronomy & Astrophysics*, 655, A106, 2021. DOI: 10.1051/0004-6361/202141740.
6. **Perraut, K., Cunha, M., Romanovskaya, A., Shulyak, D., Ryabchikova, T., et al.** Benchmarking the fundamental parameters of Ap stars with optical long-baseline interferometric measurements. *Astronomy & Astrophysics*, 642, A101, 2020.
7. **Romanovskaya, A., Ryabchikova, T., Shulyak, D., Perraut, K., Valyavin, G., Burlakova, T., & Galazutdinov, G.** Fundamental parameters and evolutionary status of magnetic chemically peculiar stars HD 188041, HD 111133, and HD 204411: spectroscopy versus interferometry. *MNRAS*, 488, 2343, 2019.

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HIGHLIGHTS

Research areas

Stellar spectroscopy; Ap/Bp, Am and normal A-B stars; spectroscopic binaries

Methods

Spectral synthesis; LTE/non-LTE abundance analysis; SED fitting; chi-square optimization; PCA; ML regression

Software

Python, NumPy, SciPy, pandas, scikit-learn, LightGBM, XGBoost, PyTorch, Optuna

PUBLICATION METRICS

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RESEARCH EXPERIENCE AND METHODS

RESEARCH EXPERIENCE

- Self-consistent modelling of stellar atmospheres and SEDs for magnetic chemically peculiar Ap/Bp stars.
- Determination of fundamental parameters, chemical abundances, magnetic-field estimates, rotational broadening parameters and stellar radii from high-resolution spectra.
- LTE and non-LTE abundance analysis of A-B stars with low rotational velocities and criteria for chemically normal stars.
- Vertical stratification analysis of Fe, Cr, Ca, Si and rare-earth elements in chemically peculiar stellar atmospheres.
- Analysis of the spectroscopic binary α Equulei and development of approaches for binary-star spectral analysis.

TECHNICAL SKILLS

- Scientific Python: NumPy, SciPy, pandas, matplotlib, Astropy, scikit-learn, Jupyter.
- Machine learning and optimization: PCA, regression models, LightGBM/XGBoost, PyTorch, Optuna, robustness analysis.
- Spectroscopy and atmosphere modelling: LLmodels, SynthV/Synmast, VALD3, SME, BinMag6, SED fitting.
- Reproducible workflows: synthetic dataset generation, pipeline design, validation on synthetic and observed spectra, documentation.

SELECTED CONFERENCE TALKS

- OBA stars: Variability and Magnetic Fields, St. Petersburg State University, 2021 - oral talk on Ap stars HD 120198 and HD 220825.
- All-Russian Astronomical Conference (VAK-2021), Moscow, 2021 - oral talk on spectroscopy and interferometry of Ap stars.
- Ground-based Astronomy in Russia. XXI century, SAO RAS, 2020 - oral talk on fundamental parameters of Ap stars.
- Physics of stars and planets: atmospheres, activity and magnetic fields, Shamakhi, Azerbaijan, 2019 - oral talk on Ap-stars 78 Vir and AX CVn.
- Physics of Magnetic Stars, Special Astrophysical Observatory, 2018 - oral talk on vertical stratification and surface chemical inhomogeneity in Ap stars.

TEACHING AND TRAINING

- Tutorial guide for PhD students: VALD3 - Vienna Atomic Line Database, Institute of Astronomy RAS.
- Professional program: Artificial Intelligence in Natural Sciences, Yandex School of Data Analysis, 2025-2027.